

```
[4]: import numpy as np
import pandas as pd

[5]: df = pd.read_csv('/home/subramanyaprasad/Machine Learning/Data Analysis/Notes/03-Pandas/tips.csv')

[6]: df

[6]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
0      16.99  1.01  Female    No  Sun  Dinner    2         8.49  Christy Cunningham  3560325168603410  Sun2959
1      10.34  1.66   Male    No  Sun  Dinner    3         3.45   Douglas Tucker  4478071379779230  Sun4608
2      21.01  3.50   Male    No  Sun  Dinner    3         7.00   Travis Walters  6011812112971322  Sun4458
3      23.68  3.31   Male    No  Sun  Dinner    2        11.84  Nathaniel Harris  4676137647685994  Sun5260
4      24.59  3.61  Female    No  Sun  Dinner    4         6.15   Tonya Carter  4832732618637221  Sun2251
...      ...    ...    ...    ...    ...    ...    ...      ...      ...      ...
239    29.03  5.92   Male    No  Sat  Dinner    3         9.68   Michael Avila  5296068606052842  Sat2657
240    27.18  2.00  Female    Yes  Sat  Dinner    2        13.59  Monica Sanders  3506806155565404  Sat1766
241    22.67  2.00   Male    Yes  Sat  Dinner    2        11.34   Keith Wong  6011891618747196  Sat3880
242    17.82  1.75   Male    No  Sat  Dinner    2         8.91   Dennis Dixon  4375220550950  Sat17
243    18.78  3.00  Female    No  Thur Dinner    2         9.39   Michelle Hardin  3511451626698139  Thur672

244 rows x 11 columns

Let's do filter based on a single condition:

Suppose we want rows with total bill more than $40:

[12]: df['total_bill'] > 40

[12]:
0      False
1       False
2       False
3       False
4       False
...
239    False
240    False
241    False
242    False
243    False
Name: total_bill, Length: 244, dtype: bool

[10]: bool_series = df['total_bill'] > 40

[11]: df[bool_series]

[11]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
59    48.27  6.73   Male    No  Sat  Dinner    4        12.07   Brian Ortiz  6596453823950595  Sat8139
95    40.17  4.73   Male    Yes  Fri  Dinner    4        10.04   Aaron Bentley  18002661638690  Fri9628
102   44.30  2.50  Female    Yes  Sat  Dinner    3        14.77   Heather Cohen  379771118886604  Sat6240
142   41.19  5.00   Male    No  Thur  Lunch    5         8.24   Eric Andrews  4356531761046453  Thur3621
156   48.17  5.00   Male    No  Sun  Dinner    6         8.03   Ryan Gonzales  3523151482063321  Sun7518
170   50.81  10.00  Female    Yes  Sat  Dinner    3        16.94   Gregory Clark  5473850968388236  Sat1954
182   45.35  3.50   Male    Yes  Sun  Dinner    3        15.12   Jose Parsons  4112207559459910  Sun2337
184   40.55  3.00   Male    Yes  Sun  Dinner    2        20.27   Stephen Cox  3547798222044029  Sun5140
197   43.11  5.00  Female    Yes  Thur  Lunch    4        10.78   Brooke Soto  5544902205760175  Thur9313
212   48.33  9.00   Male    No  Sat  Dinner    4        12.08   Alex Williamson  676218815212  Sat4590

We can do in one step

[13]: df[df['total_bill'] > 40]

[13]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
59    48.27  6.73   Male    No  Sat  Dinner    4        12.07   Brian Ortiz  6596453823950595  Sat8139
95    40.17  4.73   Male    Yes  Fri  Dinner    4        10.04   Aaron Bentley  18002661638690  Fri9628
102   44.30  2.50  Female    Yes  Sat  Dinner    3        14.77   Heather Cohen  379771118886604  Sat6240
142   41.19  5.00   Male    No  Thur  Lunch    5         8.24   Eric Andrews  4356531761046453  Thur3621
156   48.17  5.00   Male    No  Sun  Dinner    6         8.03   Ryan Gonzales  3523151482063321  Sun7518
170   50.81  10.00  Female    Yes  Sat  Dinner    3        16.94   Gregory Clark  5473850968388236  Sat1954
182   45.35  3.50   Male    Yes  Sun  Dinner    3        15.12   Jose Parsons  4112207559459910  Sun2337
184   40.55  3.00   Male    Yes  Sun  Dinner    2        20.27   Stephen Cox  3547798222044029  Sun5140
197   43.11  5.00  Female    Yes  Thur  Lunch    4        10.78   Brooke Soto  5544902205760175  Thur9313
212   48.33  9.00   Male    No  Sat  Dinner    4        12.08   Alex Williamson  676218815212  Sat4590

[14]: df[df['sex'] == "Male"]

[14]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
1      10.34  1.66   Male    No  Sun  Dinner    3         3.45   Douglas Tucker  4478071379779230  Sun4608
2      21.01  3.50   Male    No  Sun  Dinner    3         7.00   Travis Walters  6011812112971322  Sun4458
3      23.68  3.31   Male    No  Sun  Dinner    2        11.84   Nathaniel Harris  4676137647685994  Sun5260
5      25.29  4.71   Male    No  Sun  Dinner    4         6.32   Erik Smith  213140353657882  Sun9679
6      8.77  2.00   Male    No  Sun  Dinner    2         4.38  Kristopher Johnson  2223727524230344  Sun5985
...      ...    ...    ...    ...    ...    ...    ...      ...      ...      ...
236   12.60  1.00   Male    Yes  Sat  Dinner    2         6.30   Matthew Myers  3543676378973965  Sat5032
237   32.83  1.17   Male    Yes  Sat  Dinner    2        16.42   Thomas Brown  4284722681265508  Sat2929
239    29.03  5.92   Male    No  Sat  Dinner    3         9.68   Michael Avila  5296068606052842  Sat2657
241    22.67  2.00   Male    Yes  Sat  Dinner    2        11.34   Keith Wong  6011891618747196  Sat3880
242    17.82  1.75   Male    No  Sat  Dinner    2         8.91   Dennis Dixon  4375220550950  Sat17

157 rows x 11 columns

[15]: df[df['size'] > 3]

[15]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
4      24.59  3.61  Female    No  Sun  Dinner    4         6.15   Tonya Carter  4832732618637221  Sun2251
5      25.29  4.71   Male    No  Sun  Dinner    4         6.32   Erik Smith  213140353657882  Sun9679
7      26.88  3.12   Male    No  Sun  Dinner    4         6.72   Robert Buck  3514785077705092  Sun8157
11     35.26  5.00  Female    No  Sun  Dinner    4         8.82   Diane Macias  4577817359320969  Sun6886
13     18.43  3.00   Male    No  Sun  Dinner    4         4.61   Joshua Jones  6011163105616890  Sun2971
23     39.42  7.58   Male    No  Sat  Dinner    4         9.86   Lance Peterson  3542584081609808  Sat239
25     17.81  2.34   Male    No  Sat  Dinner    4         4.45   Robert Perkins  30502930499388  Sat907
31     26.38  2.50   Male    No  Sat  Dinner    4         4.59   Danny Santiago  630415546013  Sat4947
33     20.69  2.45  Female    No  Sat  Dinner    4         5.17   Amber Francis  37742985258914  Sat6649
44     30.40  5.60   Male    No  Sun  Dinner    4         7.60   Todd Cooper  503846761263  Sun2274
47     32.40  6.00   Male    No  Sun  Dinner    4         8.10   James Barnes  3552002592874186  Sun9677
52     34.81  5.20  Female    No  Sun  Dinner    4         8.70   Emily Daniel  4291280793094374  Sun6165
54     25.56  4.34   Male    No  Sun  Dinner    4         6.39   Ronald Owens  6569907991983380  Sun9470
56     38.01  3.00   Male    Yes  Sat  Dinner    4         9.50   James Christensen DDS  349793629453226  Sat8903
59    48.27  6.73   Male    No  Sat  Dinner    4        12.07   Brian Ortiz  6596453823950595  Sat8139
63     18.29  3.76   Male    Yes  Sat  Dinner    4         4.57   Chad Hart  580171498976  Sat4178
77     27.20  4.00   Male    No  Thur  Lunch    4         6.80   John Davis  30344778738589  Thur4924
85     34.83  5.17  Female    No  Thur  Lunch    4         8.71   Shawna Cook  6011787464177340  Thur7972
95     40.17  4.73   Male    Yes  Fri  Dinner    4        10.04   Aaron Bentley  18002661638690  Fri9628
116    29.93  5.07   Male    No  Sun  Dinner    4         7.48   Shawn Blake  468907971213722  Sun22
119    24.08  2.92  Female    No  Thur  Lunch    4         6.02   Melanie Jordan  676212062720  Thur8063
125    29.80  4.20  Female    No  Thur  Lunch    6         4.97   Angela Sanchez  503857080488  Thur3948
141    34.30  6.70   Male    No  Thur  Lunch    6         5.72   Steven Carlson  3526515703718508  Thur1025
142     41.19  5.00   Male    No  Thur  Lunch    5         8.24   Eric Andrews  4356531761046453  Thur3621
143    27.05  5.00  Female    No  Thur  Lunch    6         4.51   Regina Jones  4311048695487  Thur6179
153    24.55  2.00   Male    No  Sun  Dinner    4         6.14   Todd Patterson  4416804908942159  Sun8670
154    19.77  2.00   Male    No  Sun  Dinner    4         4.94   James Smith  213169731428229  Sun5814
155    29.85  5.14  Female    No  Sun  Dinner    5         5.97   Madison Wilson  4210875236164664  Sun9176
156    48.17  5.00   Male    No  Sun  Dinner    6         8.03   Ryan Gonzales  3523151482063321  Sun7518
157    25.00  3.75  Female    No  Sun  Dinner    4         6.25   Laura Robles  213158685144262  Sun7015
159    16.49  2.00   Male    No  Sun  Dinner    4         4.12   Christopher Soto  30501814271434  Sun1781
160    21.50  3.50   Male    No  Sun  Dinner    4         5.38   Travis Gonzalez  3527688419764685  Sun245
167    31.71  4.50   Male    No  Sun  Dinner    4         7.93   Michael Lawson  3566285921227119  Sun3719
180    34.65  3.68   Male    Yes  Sun  Dinner    4         8.66   James Hebert DDS  676168737648  Sun7544
183    23.17  6.50   Male    Yes  Sun  Dinner    4         5.79   Dr. Michael James  4718501859162  Sun6059
185    20.69  5.00   Male    No  Sun  Dinner    5         4.14   Joseph Howell  30362402455623  Sun5842
187    30.46  2.00   Male    Yes  Sun  Dinner    5         6.09   David Barrett  4792882899700988  Sun9987
197    43.11  5.00  Female    Yes  Thur  Lunch    4        10.78   Brooke Soto  5544902205760175  Thur9313
204    20.53  4.00   Male    Yes  Thur  Lunch    4         5.13   Scott Kim  3570811756827620  Thur2160
207    38.73  3.00   Male    Yes  Sat  Dinner    4         9.68   Ricky Ramirez  347817964484033  Sat4505
211    25.89  5.16   Male    Yes  Sat  Dinner    4         6.47   Christopher Li  6011962464150569  Sat6735
212    48.33  9.00   Male    No  Sat  Dinner    4        12.08   Alex Williamson  676218815212  Sat4590
216    28.15  3.00   Male    Yes  Sat  Dinner    5         5.63   Shawn Barnett PhD  4590982568244  Sat7320
219    30.14  3.09  Female    Yes  Sat  Dinner    4         7.54   Shelby House  502097403252  Sat8863
227    20.45  3.00   Male    No  Sat  Dinner    4         5.11   Robert Bradley  213141668145910  Sat4319
230    24.01  2.00   Male    Yes  Sat  Dinner    4         6.00   Michael Osborne  4258682154026  Sat7872

What if there are multiple conditions?

AND & --- Both conditions need to be True; OR | --- Either condition is True

[16]: df[(df['total_bill'] > 30) & (df['sex'] == "Male")]

[16]:
0      False
1       True
2       True
3       True
4      False
...
239    False
240    False
241    True
242    True
243    False
Name: sex, Length: 244, dtype: bool

Both conditions are True

[17]: df[(df['total_bill'] > 30) & (df['sex'] == "Male")]

[17]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
23    39.42  7.58   Male    No  Sat  Dinner    4         9.86   Lance Peterson  3542584061609808  Sat239
39    31.27  5.00   Male    No  Sat  Dinner    3        10.42   Mr. Brandon Berry  6011525851069856  Sat6373
44    30.40  5.60   Male    No  Sun  Dinner    4         7.60   Todd Cooper  503846761263  Sun2274
47    32.40  6.00   Male    No  Sun  Dinner    4         8.10   James Barnes  3552002592874186  Sun9677
56    38.01  3.00   Male    Yes  Sat  Dinner    4         9.50   James Christensen DDS  349793629453226  Sat8903
59    48.27  6.73   Male    No  Sat  Dinner    4        12.07   Brian Ortiz  6596453823950595  Sat8139
83    32.68  5.00   Male    Yes  Thur  Lunch    2        16.34   Daniel Murphy  535617750109133  Thur8801
95    40.17  4.73   Male    Yes  Fri  Dinner    4        10.04   Aaron Bentley  18002661638690  Fri9628
112   38.07  4.00   Male    No  Sun  Dinner    3        12.69   Jeff Lopez  3572865915178463  Sun591
141   34.30  6.70   Male    No  Thur  Lunch    6         5.72   Steven Carlson  3526515703718508  Thur1025
142   41.19  5.00   Male    No  Thur  Lunch    5         8.24   Eric Andrews  4356531761046453  Thur3621
156   48.17  5.00   Male    No  Sun  Dinner    6         8.03   Ryan Gonzales  3523151482063321  Sun7518
167   31.71  4.50   Male    No  Sun  Dinner    4         7.93   Michael Lawson  3566285921227119  Sun3719
170   50.81  10.00  Female    Yes  Sat  Dinner    3        16.94   Gregory Clark  5473850968388236  Sat1954
173   31.85  3.18   Male    Yes  Sun  Dinner    2        15.92   Scott Perez  3577115550328507  Sun9335
175   32.90  3.11   Male    Yes  Sun  Dinner    2        16.45   Nathan Reynolds  370307040837149  Sun5109
179   34.63  3.55   Male    Yes  Sun  Dinner    2        17.32   Brian Bailey  346656312114848  Sun9851
180   34.65  3.68   Male    Yes  Sun  Dinner    4         8.66   James Hebert DDS  676168737648  Sun7544
182   45.35  3.50   Male    Yes  Sun  Dinner    3        15.12   Jose Parsons  4112207559459910  Sun2337
184   40.55  3.00   Male    Yes  Sun  Dinner    2        20.27   Stephen Cox  3547798222044029  Sun5140
187   30.46  2.00   Male    Yes  Sun  Dinner    5         6.09   David Barrett  4792882899700988  Sun9987
207   38.73  3.00   Male    Yes  Sat  Dinner    4         9.68   Ricky Ramirez  347817964484033  Sat4505
210   30.06  2.00   Male    Yes  Sat  Dinner    3        10.02   Shawn Mendoza  30184049218122  Sat8361
212   48.33  9.00   Male    No  Sat  Dinner    4        12.08   Alex Williamson  676218815212  Sat4590
237   32.83  1.17   Male    Yes  Sat  Dinner    2        16.42   Thomas Brown  4284722681265508  Sat2929

Either condition is True:

[19]: df[(df['total_bill'] > 30) | (df['sex'] == "Male")]

[19]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
1      10.34  1.66   Male    No  Sun  Dinner    3         3.45   Douglas Tucker  4478071379779230  Sun4608
2      21.01  3.50   Male    No  Sun  Dinner    3         7.00   Travis Walters  6011812112971322  Sun4458
3      23.68  3.31   Male    No  Sun  Dinner    2        11.84   Nathaniel Harris  4676137647685994  Sun5260
5      25.29  4.71   Male    No  Sun  Dinner    4         6.32   Erik Smith  213140353657882  Sun9679
6      8.77  2.00   Male    No  Sun  Dinner    2         4.38  Kristopher Johnson  2223727524230344  Sun5985
...      ...    ...    ...    ...    ...    ...    ...      ...      ...      ...
237   32.83  1.17   Male    Yes  Sat  Dinner    2        16.42   Thomas Brown  4284722681265508  Sat2929
238   35.83  4.67  Female    No  Sat  Dinner    3        11.94   Kimberly Crane  676184013727  Sat9777
239    29.03  5.92   Male    No  Sat  Dinner    3         9.68   Michael Avila  5296068606052842  Sat2657
241    22.67  2.00   Male    Yes  Sat  Dinner    2        11.34   Keith Wong  6011891618747196  Sat3880
242    17.82  1.75   Male    No  Sat  Dinner    2         8.91   Dennis Dixon  4375220550950  Sat17

164 rows x 11 columns

isin() is used to check whether each value belongs to a given list, set, tuple, or another Series.

Syntax: df['column'].isin(values)

[20]: options = ['Sat', 'Sun']

[24]: df[df['day'].isin(options)]

[24]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
0      16.99  1.01  Female    No  Sun  Dinner    2         8.49  Christy Cunningham  3560325168603410  Sun2959
1      10.34  1.66   Male    No  Sun  Dinner    3         3.45   Douglas Tucker  4478071379779230  Sun4608
2      21.01  3.50   Male    No  Sun  Dinner    3         7.00   Travis Walters  6011812112971322  Sun4458
3      23.68  3.31   Male    No  Sun  Dinner    2        11.84   Nathaniel Harris  4676137647685994  Sun5260
4      24.59  3.61  Female    No  Sun  Dinner    4         6.15   Tonya Carter  4832732618637221  Sun2251
...      ...    ...    ...    ...    ...    ...    ...      ...      ...      ...
238   35.83  4.67  Female    No  Sat  Dinner    3        11.94   Kimberly Crane  676184013727  Sat9777
239    29.03  5.92   Male    No  Sat  Dinner    3         9.68   Michael Avila  5296068606052842  Sat2657
240    27.18  2.00  Female    Yes  Sat  Dinner    2        13.59  Monica Sanders  3506806155565404  Sat1766
241    22.67  2.00   Male    Yes  Sat  Dinner    2        11.34   Keith Wong  6011891618747196  Sat3880
242    17.82  1.75   Male    No  Sat  Dinner    2         8.91   Dennis Dixon  4375220550950  Sat17

163 rows x 11 columns

[25]: df[df['day'].isin(['Sat', 'Sun'])]

[25]:
total_bill  tip    sex  smoker  day    time  size  price_per_person  Payer Name  CC Number  Payment ID
0      16.99  1.01  Female    No  Sun  Dinner    2         8.49  Christy Cunningham  3560325168603410  Sun2959
1      10.34  1.66   Male    No  Sun  Dinner    3         3.45   Douglas Tucker  4478071379779230  Sun4608
2      21.01  3.50   Male    No  Sun  Dinner    3         7.00   Travis Walters  6011812112971322  Sun4458
3      23.68  3.31   Male    No  Sun  Dinner    2        11.84   Nathaniel Harris  4676137647685994  Sun5260
4      24.59  3.61  Female    No  Sun  Dinner    4         6.15   Tonya Carter  4832732618637221  Sun2251
...      ...    ...    ...    ...    ...    ...    ...      ...      ...      ...
238   35.83  4.67  Female    No  Sat  Dinner    3        11.94   Kimberly Crane  676184013727  Sat9777
239    29.03  5.92   Male    No  Sat  Dinner    3         9.68   Michael Avila  5296068606052842  Sat2657
240    27.18  2.00  Female    Yes  Sat  Dinner    2        13.59  Monica Sanders  3506806155565404  Sat1766
241    22.67  2.00   Male    Yes  Sat  Dinner    2        11.34   Keith Wong  6011891618747196  Sat3880
242    17.82  1.75   Male    No  Sat  Dinner    2         8.91   Dennis Dixon  4375220550950  Sat17

163 rows x 11 columns

[ ]:
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